

 MOTION IMAGERY STANDARDS BOARD STANDARD Constructs to Amend/Segment KLV Metadata	MISB ST 1607.2 2025-06-11
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1 Scope

This standard defines two KLV Local Set constructs per the definitions in SMPTE ST 336 [1] to amend and segment KLV metadata.

A metadata life cycle begins at the creation of metadata called first-generation metadata. This life cycle includes adding additional metadata, editing existing metadata and deleting metadata in subsequent processing. One of the goals of the MISB is to retain all metadata – both first-generation metadata and subsequent changes in metadata, for analysis and legal purposes.

The standard defines the Amend Local Set for editing metadata using a Tree structure of parent/child relationships, while preserving first-generation metadata. The Amend Local Set enables child or branch metadata, residing within the first-generation metadata’s set (the parent or root metadata set), and provides tight cohesion between first-generation and changed metadata. The Amend Local Set does not have predefined member metadata items; it “inherits” members from the parent metadata sets definition (e.g., a Local Set’s Standard) based on application needs.

This standard also defines the Segment Local Set for distributing metadata between an instantiating parent set, and one or more child sets. The Segment Local Set does not have predefined member metadata items; it “inherits” members from the parent metadata set’s definition based on application needs.

The standard also defines requirements for using the Amend Local Set and Segment Local Set within a root metadata set.

2 References

- [1] SMPTE, *ST 336:2007 - Data Encoding Protocol Using Key-Length-Value*.
- [2] MISB, *MISP-2025.1: Motion Imagery Handbook*, 2025-06-11.
- [3] MISB, *ST 0807.27 - MISB KLV Metadata Registry*, 2025-06-11.
- [4] MISB, *ST 0102.12 - Security Metadata Universal and Local Sets for Motion Imagery Data*, 2017-06-22.
- [5] MISB, *MISP-2025.1: Motion Imagery Standards Profile*, 2025-06-11.

3 Revision History

Version	Approved Date	Change Description
ST 1607.2	2025-06-11	- Added new requirements -07, -08 & -09 - Retired requirement -03 & added Appendix A - Editorial revisions to requirements -01, -02, -04, -05 & -06 - Revised Section titles to align to latest MISB document template - Revised Section 9 and added Sections 10 & 11 to aid/clarify implementation rules

4 Definitions of Terms, Acronyms, and Initialisms

4.1 Terms

The following lists the terms and their definitions this document uses.

Element	Denotes a SMPTE ST 336 data item or data group
First-generation metadata	The initial creation of metadata at its source; constitutes the root level metadata within a metadata tree hierarchy

4.2 Acronyms and Initialisms

The following lists the acronyms and initialisms this document uses.

KLV	Key Length Value
LS	Local Set
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
MSID	Metadata Substream Identifier
SMPTE	Society of Motion Picture and Television Engineers
ST	Standard

5 Introduction

New user applications often need to define new metadata elements. Beyond adding new metadata elements, however, is the need to extend metadata, such as reusing one or more metadata elements per instantiation of a metadata set, and editing existing metadata while preserving first-generation metadata.

The Motion Imagery Handbook [2] Section 7.1 introduces the notion of a metadata life cycle, where data preservation and data sharing are key concepts in the management of metadata. As introduced in the handbook, the enhanced operations use “Amend Metadata” to modify existing metadata while preserving original data, and “Segment Metadata” to share metadata elements while partitioning others to describe related but different information. Both operations meet the goals of data preservation and data sharing. Tree structures built using Amend Metadata and Segment Metadata form parent/child relationships, where the top-most parent is the “root”, and a

child is a “branch” of the tree. For implementation, the “Amend Local Set” (i.e., Amend LS) and the “Segment Local Set” (Segment LS) enable data preservation and data sharing, respectively.

The Amend LS and Segment LS constructs are somewhat different from past uses of Local Sets. Within the SMPTE ST 336 document defining KLV, coding of a KLV triplet within the data Value of KLV requires a category designator called groups, which include sets and packs. Both constructs will – by design – contain Tag-Length-Value (TLV) triplets in their Value; thus, the Amend and Segment constructs are Local Sets. However, unlike other MISB Local Sets, the Segment LS and Amend LS have no predefined data elements.

In their use, an instantiating metadata set provides the source of metadata data elements. The Segment LS and Amend LS inherit metadata elements from their parent set based on application. When added to an instantiating MISB metadata set, these KLV constructs provide new functionality, such as metadata editing, metadata preservation and metadata sharing, thereby extending the instantiating metadata set to meet new application needs.

5.1 Metadata Substream Identifier (MSID)

A Metadata Substream Identifier (MSID) provides a unique identifier to each Amend LS or Segment LS. This allows for multiple instances of these LS within a metadata stream in support of sub-streams of sub-streams and Report-on-change. See the Motion Imagery Handbook for an in-depth discussion of the sub-stream construction and the necessity of including the MSID.

An instantiating metadata set defines a MSID metadata item and its requirements for its use. Every Amend/Segment LS implemented within that instantiating metadata set then includes that MSID as a metadata item.

6 Amend Local Set

The Amend Local Set enables editing of first-generation metadata, while preserving first-generation metadata. Using a tree hierarchical model, first-generation metadata lies at the root level. One or more child (branch) sub-streams contain any changes to first-generation metadata as shown in Figure 1. This model allows construction of multiple parent/child relations.

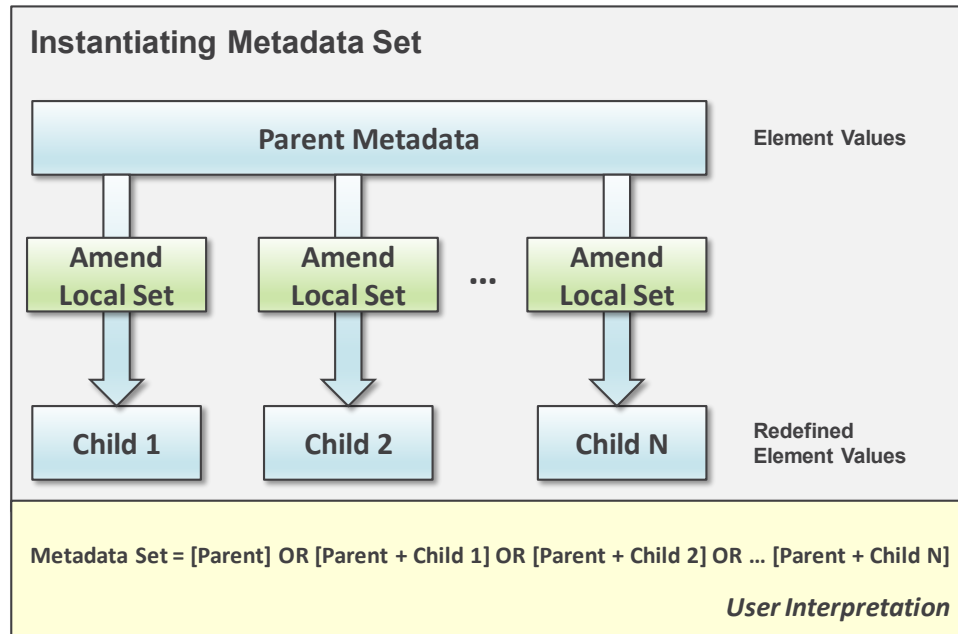


Figure 1: Parent/Child levels of metadata using the Amend Local Set.

First-generation metadata remains unaltered at the root level and thus preserved. An Amend child set can indicate the addition, deletion, or value change of metadata elements in the instantiating parent set (The Motion Imagery Handbook describes these operations). The user can interpret the metadata in two ways, either as the original first-generation metadata, or a combination of first-generation metadata plus any edits of metadata within the child set.

As an example, assume a process operates on the first-generation metadata resulting in more accurate metadata element values. Since applications must preserve first-generation metadata, an Amend LS allows for supplying additional – possibly more accurate metadata values while leaving the first-generation metadata intact. Multiple uses of the Amend LS introduce multiple child sets, as Figure 1 illustrates.

Figure 2 illustrates a parent set (in this case the Root LS) and an Amend LS with updates. Parent metadata Elements with Items A, B and C are the first-generation metadata. Item Z is an Amend LS child with metadata elements A', B' and C', each having different values for Items A, B and C. Although the item identifiers are the same, the elements in the child level have different values than those in the parent level. Users may choose which metadata to use (e.g., view); either Items A, B, C or Items A', B', C'.

The Amend LS includes the MSID (Item M) to distinguish this child (Item Z) from a different child (i.e., a different Item Z). The root LS document must define a MSID item in its LS whenever the root LS allows for Amend LSs.

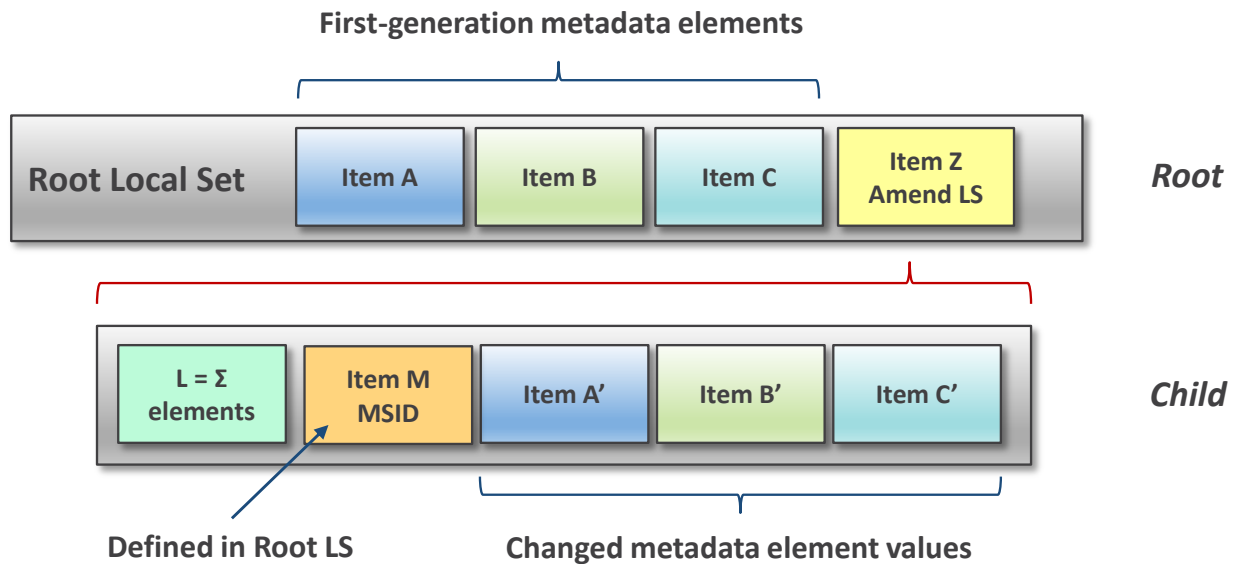


Figure 2: Parent/Child example – Amend LS.

Although this example has only one child, reusing the Amend LS multiple times within a metadata set allows multiple instances of changes to first-generation metadata, thus constructing multiple child levels. Nesting within a child forms a sub-child relationship.

MISB ST 0807 [3] registers the Amend Local Set 16-Byte Universal Label as:

06.0E.2B.34.02.0B.01.01.0E.01.03.03.03.01.00.00 (CRC 17182)

7 Segment Local Set

The Segment Local Set defines “branches” of independent metadata, while sharing a common set of metadata from the parent set. A parent set can embed one or more Segment LSs defining parallel sets (branches) of metadata. Each Segment LS signals a “union and override” of metadata elements with the list of parent metadata elements. The “union and override” means the list of elements in the Segment LS combine with the list of elements from the parent set forming a new set for users to interpret. However, when there are duplicates, the elements in the Segment LS take precedence in the resulting set, i.e., they override the elements from the parent set.

Using one or more Segment LSs reuses or shares a parent’s metadata elements with each child Segment LS. Figure 3 illustrates an example parent set with three Segment LSs. The parent set provides common elements each Segment LS will union and override. The bottom of the figure shows how the user needs to interpret the result, and in this case, there are three independent resulting metadata sets.

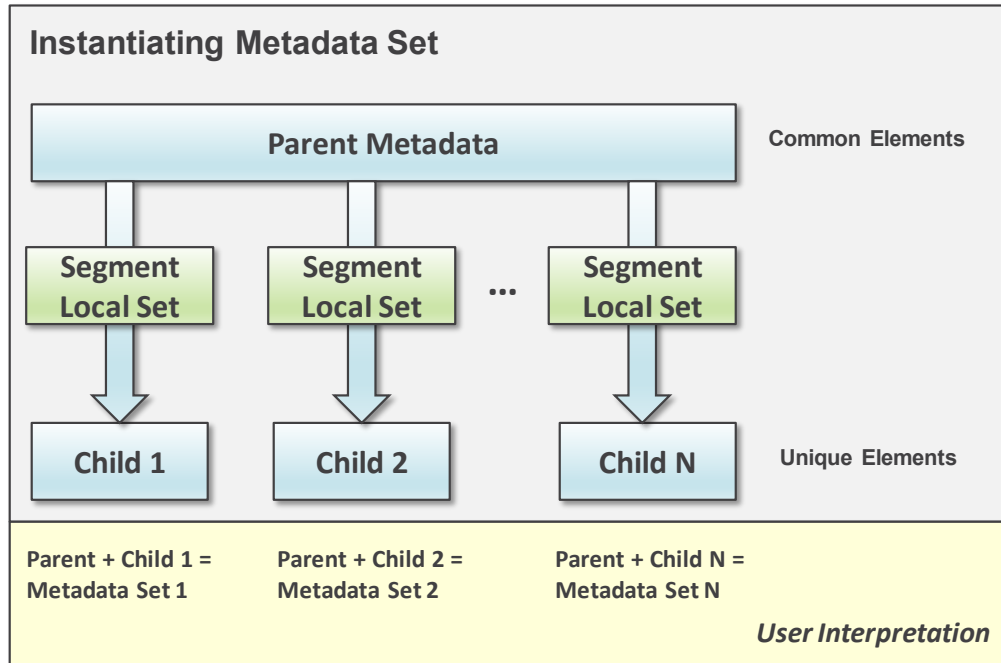


Figure 3: Parent/Children level branching using the Segment Local Set.

As an example, assume the formatting of two images from two sensors on the same platform into one composite image. Each sensor shares common metadata regarding the platform (i.e., elevation, position, etc.), while each sensor has unique metadata (i.e., MIIS, sensor type, etc.). The unique metadata elements override the same element identifiers in the parent set but with different values.

To illustrate the use of the Segment LS, in Figure 4 the parent (i.e., Root LS in this case) defines metadata elements Items A, B and C.

The Segment LS includes new values for parent metadata elements Items A' and C', and the Segment LS adds the metadata element Item F. All elements at the child level have values which are independent of those at the parent level. The resulting metadata set is a “union and replace” of the parent metadata elements with the Segment LS metadata elements.

The Segment LS includes the MSID (Item M) to distinguish this child (Item X) from a different child (i.e., a different Item X). The root LS document must define a MSID item in its LS whenever the root LS allows for Segment LSs.

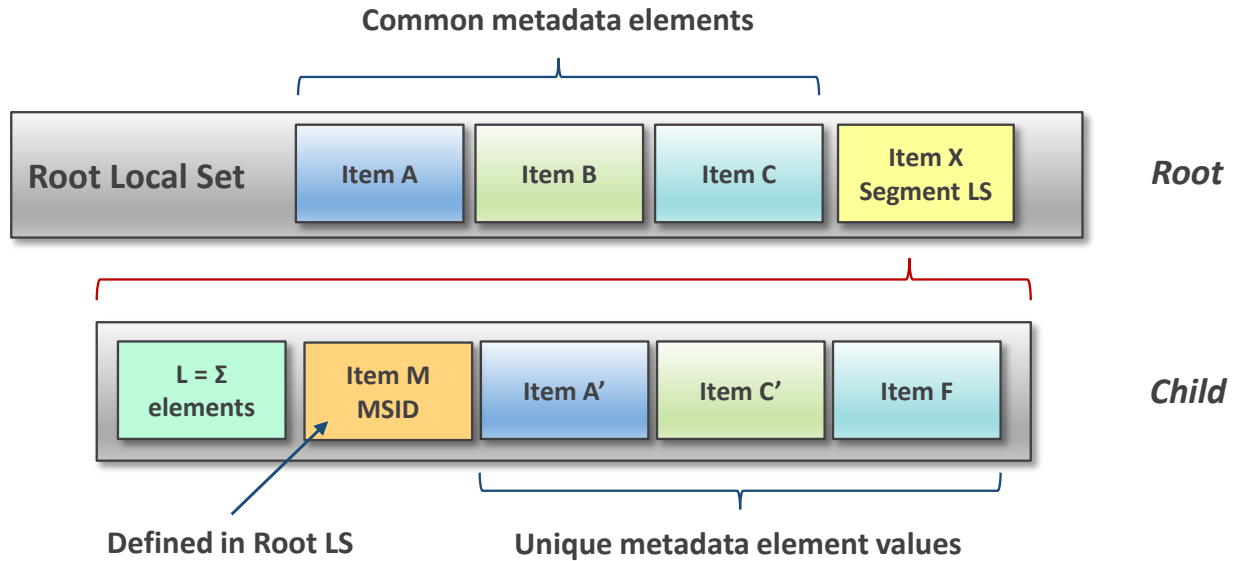


Figure 4: Parent/Child example – Segment LS.

MISB ST 0807 registers the Segment LS 16-byte Universal Label as:

06.0E.2B.34.02.0B.01.01.0E.01.03.03.03.00.00.00 (CRC 29742)

8 Limitations

Certain use cases in using the Amend LS and Segment LS, particularly the Amend LS, may present issues as described below.

- **Duplicated metadata elements (i.e., items) in a parent set:** When multiple instances of the same metadata element appear in a parent metadata set, there is no defined mechanism to discern which metadata elements within an Amend LS or Segment LS relate to a specific instance in the parent set. This is not an issue when ST 0601 is the parent set, since duplicate elements are not likely; however, use of these constructs in other MISB metadata sets in the future are likely.
- **Ordered metadata elements in a parent set:** In metadata sets with KLV pack structures, certain packs mandate a specific ordering of their elements. In reusing such a pack in a child set, preserving the element ordering is important.

Therefore, exercise caution in employing these constructs in consideration of their intended scope and resulting interpretation. This document will include guidance should these issues arise.

9 Rules Governing Amend and Segment Local Set Security Items

MISB ST 0102 [4] requires the root level metadata set to have a complete security set and represent the overall classification of the data within the stream. Where a KLV metadata set includes one or more Amend LSs and/or Segment LSs, the root level security metadata set indicates the overall classification of all Amends/Segments and their children.

ST 1607.2 Constructs to Amend/Segment KLV Metadata

Two metadata items within ST 0102, Object Country Coding Method (Item 12) and Object Country Codes (Item 13), may be present within a child level if needed. For cases where all data within a stream pertains to the same country, specify the Object Country Coding Method and Object Country Codes at the root level.

Requirement(s)	
ST 1607-01	Where a KLV metadata set includes an Amend LS, the security metadata for the root KLV metadata set shall apply to all the metadata items in the metadata set, children and sub-children.
ST 1607-02	Where a KLV metadata set includes a Segment LS, the security metadata for the root KLV metadata set shall apply to all metadata items in the metadata set, children and sub-children.
ST 1607.2-09	Where metadata within an Amend LS indicates a security set object country code unique to that Amend LS, the Amend LS shall contain a MISB ST 0102 security metadata Local Set with only Item 12 and Item 13.
ST 1607-04	Where metadata within a Segment LS indicates a security set object country code unique to that Segment LS, the Segment LS shall contain a MISB ST 0102 security metadata Local Set with only Item 12 and Item 13.

10 Rules Governing the co-use of Amend and Segment Local Sets

Embedding one or more levels of Amend and/or Segment local sets into a root set can create very complex and unusual metadata hierarchies. Amend and Segment local sets each provide unique metadata editing capabilities; however, mixing the two methods within the same parent local set creates difficulties for readers to interpret correctly. To reduce complexity, each metadata level must only use one type of editing local set, either Amend LS or Segment LS but not both. Furthermore, writers must not embed a Segment LS within an Amend LS; however, systems may embed an Amend LS within a Segment LS.

In summary, the following list outlines the rules governing the Amend LS and Segment LS:

- An Amend LS and a Segment LS may not exist simultaneously at any one level
- Multiple (child) Amend LSs may exist at any level
 - Each Amend LS child may nest other Amend LSs to desired level (Amend child within Amend child)
- An Amend LS may not nest any Segment LS
- Multiple (child) Segment LSs may exist at any level
 - Each Segment LS child may nest Segment LSs to desired level (Segment child within Segment child)
 - Each Segment LS child may nest Amend LSs to desired level (Amend Child within Segment Child)

ST 1607.2 Constructs to Amend/Segment KLV Metadata

Requirement(s)	
ST 1607.2-07	Within a given level only Amends LS(s) or Segments LS(s), but not both simultaneously, shall be allowed.
ST 1607.2-08	Where a parent Amend LS is instantiated, only child Amend LS(s) shall be allowed within the parent.

11 Rules for Amend and Segment Local Sets Conforming to the MISP

Any Motion Imagery stream created using distributed metadata elements resulting from building hierarchical parent/child metadata constructs must conform to the MISP [5].

Requirement(s)	
ST 1607-05	Where a KLV metadata set includes an Amend Local Set, the root metadata set shall contain all metadata items required to meet MISP conformance.
ST 1607-06	Where a KLV metadata set includes a Segment Local Set, the union of root, child and sub-child metadata sets shall contain all metadata items required to meet MISP conformance.

Appendix A – Retired Requirements

Requirement(s)	
ST 1607-03 (Retired)	Where a Segment Local Set is included in a KLV metadata set, the MISB ST 0102 security metadata Local Set elements Item 1, Item 2, Item 3, Item 4, Item 5, Item 6, and Item 22 (or equivalent), shall be present in the root level of the KLV metadata set.